Chair of Mathematical Statistics
Department of Mathematics
Technical University of Munich

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Statistics for Business Administration

Exam:	MA9712 / Final	**Date:**	Saturday 29th July, 2023
Examiner:	Prof. Mathias Drton	**Time:**	10:30 – 12:00

Working instructions

- This exam consists of **14 pages** with a total of **8 problems**.
  Please make sure now that you received a complete copy of the exam.
- The total amount of achievable credits in this exam is 42 credits.
- Detaching pages from the exam is prohibited.
- Allowed resources:
  - one **scientific pocket calculator**
  - the **cheat sheet**, which is contained in this exam on page **13**
- Give a reason for each answer unless explicitly stated otherwise in the respective subproblem.
- Additional space for solution to **any** of the problems is given on page 8, 12 and 14.
- Do not write with red or green colors nor use pencils.
- Physically turn off all electronic devices, put them into your bag and close the bag.

Left room from ________	to ________	/	Early submission at ________

Problem 1 (7 credits)

Give a short answer to each of the following sub-problems.

0 1

a) A study that surveyed a random sample of otherwise healthy high school students found that they are more likely to get muscle cramps when stressed. The study also noted that students drink more coffee and sleep less when they are stressed. What type of study is this? Can this study be used to conclude a causal relationship between increased stress and muscle cramps? Justify your answer.

This is an observational study.No, since the study is observational, we cannot infer causation.

0 1

b) A researcher would like to study the effect of eating breakfast on a cognitive function. Volunteers are recruited through the study by posting flyers on campus. He randomly assigns subjects to two groups, one told to eat breakfast before participating in the cognitive test and one asked to eat breakfast after the test. However, he suspects that people's usual breakfast habits (whether or not they usually eat breakfast at all) may influence his results. In order to address this, what should he do prior to assigning subjects to experimental groups?

He should first group individuals based on their usual breakfast habits into blocksand then randomise cases within each blockto the treatment groups (breakfast before/after the test).

0 1

c) A statistics student who is curious about the relationship between the amount of time students spend on social networking sites and their performance at university decides to conduct a survey. He decides to give out the survey only to his friends and asks them to fill out the survey as soon as possible. Name the type of the sample and briefly explain why the sampling approach might lead to biased results.

Convenience sample. His sample may not be representative of the population since it consists only of his friends. It is also possible that the study will have non-response bias if some choose to not bring back the survey.

0 1 2

d) Each row in the table below is a proposed grade probability distribution for the grades of a class. Identify each invalid probability distribution, and explain your reasoning.

Grades	1	2	3	4	5
P_a	0.3	0.3	0.3	0.2	0.1
P_b	0	0	1	0	0
P_c	0.3	0.3	0.3	0	0
P_d	0.3	0.5	0.2	0.1	-0.1
P_e	0.2	0.4	0.2	0.1	0.1
P_f	0	-0.1	1.1	0	0

P_a invalid. The sum of the probabilities is greater than 1 (0.3 + 0.3 + 0.3 + 0.2 + 0.1 = 1.2).
P_c invalid. The sum of the probabilities is less than 1 (0.3 + 0.3 + 0.3 + 0 + 0 = 0.9).
P_d invalid. There is a negative probability listed (-0.1 for 5).
P_f invalid. There is a negative probability listed (-0.1 for 2).

e) Suppose we have n independent random variables $X_1, X_2, \dots, X_n$, such that $\mathrm{E}(X_i) = \mu$ and $\mathrm{Var}(X_i) = \sigma^2$ for all $i \in \{1, \dots, n\}$. Calculate the variance of the random variable $\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$.

The independence of the random variables implies that they are uncorrelated, i.e., $\mathrm{Cov}(X_i, X_j) = 0 \quad \forall i, j$ with $i \neq j$. Hence, we get

$$\mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\mathrm{Var}\left(\sum_{i=1}^{n} X_i\right) = \frac{1}{n^2}\sum_{i=1}^{n}\mathrm{Var}(X_i) = \frac{1}{n^2}\sum_{i=1}^{n}\sigma^2 = \frac{1}{n^2}\cdot n \cdot \sigma^2 = \frac{\sigma^2}{n}.$$

Problem 2 (3 credits)

Jose uses the parking lot at the train station Olympiazentrum every Thursday evening. However, some days the parking lot is full, often due to events in the nearby Olympiapark. There are cultural events on 34% of evenings, sporting events on 28% of evenings, and no events on 38% of evenings. When there is a cultural event, the parking lot fills up about 22% of the time, and it fills up 67% of evenings with sporting events. On evenings when there are no events, it only fills up about 11% of the time.

If Jose comes to the parking lot and finds it full, what is the probability that there is a sporting event? Write down all steps, which are needed to compute the wanted probability. In case you use a tree, properly label the nodes in the tree.

The outcome of interest is whether there is a sporting event (we call this event A_1), and the condition is that the parking lot is full, which we denote by B. Let A_2 represent an cultural event and A_3 represent there being no event. Then the given probabilities can be written as

$$P(A_1) = 0.28 \quad P(A_2) = 0.34 \quad P(A_3) = 0.38$$
$$P(B|A_1) = 0.67 \quad P(B|A_2) = 0.22 \quad P(B|A_3) = 0.11$$

Bayes' Theorem can be used to compute the probability of a sporting event (A_1) under the condition that the parking lot is full (B):

$$\begin{aligned} P(A_1|B) &= \frac{P(B|A_1)\cdot P(A_1)}{P(B|A_1)P(A_1) + P(B|A_2)P(A_2) + P(B|A_3)P(A_3)} \\ &= \frac{0.67\cdot 0.28}{0.67\cdot 0.28 + 0.22\cdot 0.34 + 0.11\cdot 0.38} = 0.6166995 \end{aligned}$$

Problem 3 (4 credits)

The World Bank reports that 1.7% of the US population lives on less than \$2 per day. A policy maker claims that this number is misleading because of variation from state to state and rural to urban. To investigate this, she takes a random sample of 100 individuals in Atlanta to compare with the national average and finds that 4% of the Atlanta population live on less than \$2 per day.

0 1 a) The policy maker wants to show that the percentage for Atlanta is significantly higher than the national average. How should she set up her null and her alternative hypothesis?

The null and alternative hypothesis are given by

$$H_0 : p = 0.017 \qquad H_A : p > 0.017.$$

0 1 2 3 b) Can she draw her desired conclusion at a significance level of $\alpha = 0.05$? Justify your answer.

Hint:

```
pbinom(0:5, 100, prob = 0.017)
## [1] 0.1800329 0.4913817 0.7579127 0.9084860 0.9716334 0.9926011
```

In the given sample 4 out of 100 live on less than \$2 per day. Given that H_0 is true, the probability that out of 100 people, 4 or more live on less than \$2 per day is equal to

$$P(X \geq 4) = 1 - P(X \leq 3) = 0.091514,$$

where $X \sim \text{Bin}(100, 0.017)$. This probability is nothing else than the p-value of the test.Hence, we have to conclude that H_0 can't be rejected at the significance level 0.05. The data does not show strong evidence in favour of a higher percentage in Atlanta compared to the national average.

Problem 4 (8 credits)

The American National Election Studies (ANES) collects data on voter attitudes and intentions as well as demographic information. In this question we will focus on two variables from the 2012 ANES dataset:

- region (levels: Northeast, North Central, South, and West),
- whether the respondent feels things in the country are generally going in the right direction or things have pretty seriously gotten off on the wrong track.

To keep calculations simple we will work with a random sample of 400 respondents from the ANES dataset. The distribution of responses is as follows:

```
addmargins(table(anes))
##                   region
## direction          north_central northeast south west Sum
##   right_direction             38        19    46   31 134
##   wrong_track                 59        44   107   56 266
##   Sum                         97        63   153   87 400
```

According to the 2010 Census, 18% of US residents live in the Northeast, 22% live in the North Central region, 37% live in the South, and 23% live in the West.
Given the data, the following statistical analysis was done in R.

```
chisq.test(table(anes$region), p =c(0.22, 0.18, 0.37, 0.23))
##
##  Chi-squared test for given probabilities
##
## data:  table(anes$region)
## X-squared = 2.4861, df = 3, p-value = 0.4778
```

a) State the null and alternative hypothesis corresponding to the test done in the statistical analysis of the data.

0 1

We want to compare the sample distribution with the census distribution. This means, that we want to conduct a χ^2 GOF test. The corresponding hypotheses are
H_0 : The sample distribution of regions follows the census distribution.
H_A : The sample distribution of regions does not follow the census distribution.

b) Check the conditions of the test done in the statistical analysis.

0 1 2

We have to check the following conditions:
Independence: The sample is random, hence we can assume that one individual in the sample is independent of another.
Sample size: The expected counts can be calculated as follows. We take the data from the 2010 Census and multiply it by the sample size `n`

```
prob <-c(0.22, 0.18, 0.37, 0.23)
n*prob
## [1]  88  72 148  92
```

All expected counts are greater than 5, so the condition is fulfilled.

c) State the conclusion of the test in the context of the data. Justify your answer.

0 1

The test statistic is rather small, which makes the p-value rather large and in particular larger than α. We therefore fail to reject H_0.The data do not provide convincing evidence that the sample distribution of regions does not follow the census distribution.

d) Now assume we want to test the following hypotheses:

0 1 2

H_0 : Region and opinion on direction are independent.
H_A : Region and opinion on direction are dependent.

Complete the R code on the next page so that this command can be used to test the above hypotheses using the `infer` workflow.

```
library(infer)
null_dist <- anes |>

  specify(formula = __________________________________) |>

   hypothesize(null = "___________________________") |>

   _____________(reps = 1000) |>

   calculate(stat = "_________________")
```

specify(formula = direction ∼ region)

hypothesize(null = "independence")

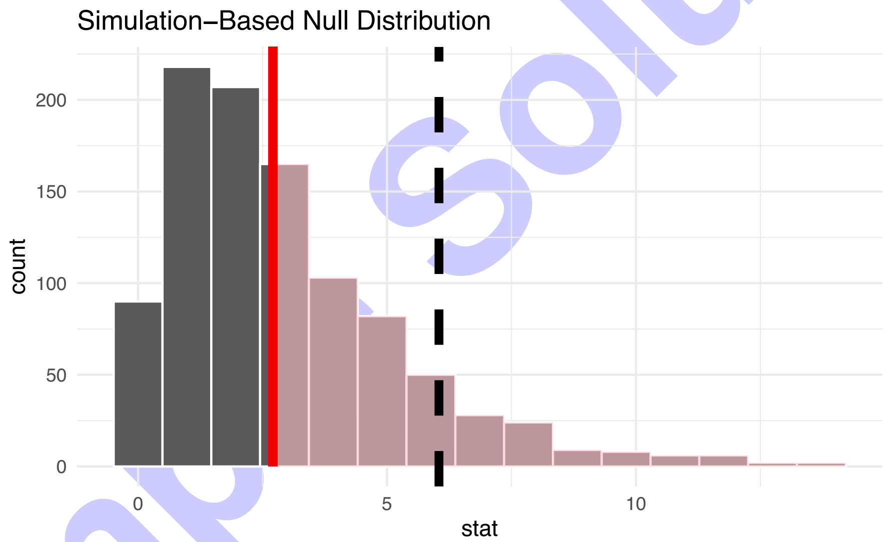
generate(reps = 1000)

calculate(stat = "Chisq")

0 1 2

e) Applying the infer workflow leads to the following simulated null distribution and an observed value for the test statistic equal to 2.7058648

```
qstat <- quantile(null_dist$stat, probs = 0.9)
null_dist |> infer::visualize() +
  shade_p_value(obs_stat = observed_stat, direction = "greater") +
  geom_vline(xintercept = qstat, size = 2, linetype = 2)
```



Given these values, decide if the null hypothesis can be rejected at a significance level of 0.1. Justify your answer by using a precise argument.

The figure shows the simulated null distribution and the shaded area corresponds to the p-value. The area is obviously large, but we can't say if it is larger than the significance level without taking further information into account. The dotted line shows the 0.9 quantile of the null distribution.This means, right to the dotted line we have an area of 0.1.Hence, the shaded area is larger than 0.1 and we fail to reject the null hypothesis.The data do not show enough evidence for the alternative of dependence between the region and the opinion about the direction.

Problem 5 (4 credits)

For a t-test in R, we obtained the following output.

```
t = 0.2024, df = 14, p-value = 0.8425
alternative hypothesis: true mean is not equal to 20
95 percent confidence interval:
18.91483 21.31131
sample estimates:
mean of x
20.11307
```

Decide if the following statements are true or false. Justify your **answer**.

Remark: Only correct justifications are awarded with points. Not the decision (true/false).

a) The null hypothesis tested here is $H_0 : \mu = 20.11307$.

0 1

False: The null hypothesis tested here is $H_0 : \mu = 20$.

b) The following R-code `pt(0.2024, df=14)` will result in an output of `0.8425`.

0 1

False: The probability $P(|T| > 0.2024)$ is equal to 0.8425.Therefore, it is the R-code `2* pt(0.2024, df=14, lower.tail = FALSE)` that will result in an output of `0.8425`.

c) The margin of error of the computed confidence interval is equal to 1.19824.

0 1

True: The given confidence interval is defined through $\overline{x} \pm ME$. Hence, $ME = 21.31131 - 20.11307 = 1.19824$

d) [18.97138, 21.25476] is a 99% confidence interval for the true mean.

0 1

False: The interval [18.97138, 21.25476] is shorter than the given 95% confidence interval. But to keep the 99% level, the 99% confidence interval for the true mean has to be wider.

Problem 6 (3 credits)

A medical research group recruits people to complete a series of short surveys about their medical history. For example, one survey asks for information on a person's family history regarding cancer. Another survey asks questions about the person's last visit to a hospital. So far, as people sign up, the average number of surveys a person completes is only 4; the standard deviation of the number of surveys is about 2.2. The research group thus wants to try a new interface that they think will encourage new enrollees to complete more surveys. To this end, they plan an experiment in which they randomize each enrollee to either the new interface or the current interface.

0 1 2 3

How many new enrollees do they need for each interface to detect an effect size of 1 additional survey being completed per enrollee? To answer this question work with a significance level of 0.05, and a desired power of 70%.

Hint:

```
qnorm(c(0.7, 0.8, 0.9, 0.95, 0.975, 0.99))
## [1] 0.524 0.842 1.282 1.645 1.960 2.326
```

We care about a difference of 1, i.e. we have an effect size $\delta_0 = 1$.
The desired power $1-\beta$ is 0.7. Together with the assumption about a standard deviation σ_0 of 2.2 in each group and the quantiles

```
qnorm(1-alpha/2)
## [1] 1.96
```

```
qnorm(power)
## [1] 0.524
```

we get

$$n \geq \frac{2\sigma_0^2}{\delta_0^2}(z_{0.975} + z_{0.7})^2 = \frac{2 \cdot 2.2^2}{1^2}(1.96 + 0.52)^2 = 59.536$$

We will need 60 new enrolleesfor each interface to detect an effect size of 1 survey per enrollee at significance level 0.05 with the power of 0.7.

Remark: One could argue that they want to test a one-sided alternative. Then they use the quantile `qnorm(1-alpha)` (=1.645) and need a sample size for each interface of

$$n \geq \frac{2\sigma_0^2}{\delta_0^2}(z_{0.95} + z_{0.7})^2 = \frac{2 \cdot 2.2^2}{1^2}(1.645 + 0.52)^2 = 45.372\,,$$

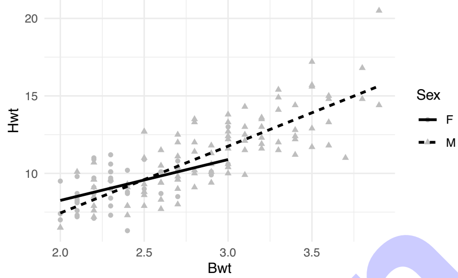
so at least 46.

Additional space for solutions–clearly mark the (sub)problem your answers are related to and strike out invalid solutions.

Problem 7 (6 credits)

The following output is a summary of a linear regression model for predicting the heart weight `Hwt` (in g) of cats from their body weight `Bwt` (in kg) and their `Sex` (M/F). The coefficients are estimated using a dataset of 144 domestic cats.

```
# A tibble: 4 × 5
  term        estimate std.error statistic  p.value
  <chr>          <dbl>     <dbl>     <dbl>    <dbl>
1 (Intercept)     2.98     1.84       1.62 0.108
2 Bwt             2.64     0.776      3.40 0.000885
3 SexM           -4.17     2.06      NA    NA
4 Bwt:SexM        1.68     0.837     NA    NA
```



a) Write out the estimated linear model for a male cat. Simplify the formula as much as possible.

0 1

The estimated linear model for a male cat is given by

$$\widehat{\text{Hwt}}_i = 2.98 + 2.64 \cdot \text{Bwt}_i - 4.17 \cdot 1 + 1.68 \cdot \text{Bwt}_i \cdot 1 = -1.19 + 4.32 \cdot \text{Bwt}_i$$

b) Interpret the slope estimate of `Sex`.

0 1

Given all else equal, the heart weight of male cats is on average reduced by 4.17 gram compared to female cats.
Remark: This reduction is (partly) compensated by the higher slope of Bwt for male cats. For a Bwt larger than $\frac{4.17}{1.68}$, the average Hwt of male cats is actually higher than that of female cats. So, only for cats with zero Bwt (they don't exists) the average Hwt of male cats will be 4.17 gram lower.

c) Decide if the interaction effect is significant at the significance level 0.05. Justify your answer by approximating the p-value using one of the values given in the hint. Give an interpretation of the result in the context of the data.

0 1 2 3 4

Hint:

```
pt(c(1.656, 1.978), df = 140, lower.tail = FALSE)
## [1] 0.0500 0.0249
```

Given the estimate for the interaction effect and corresponding standard error, we can compute the value of the test statistic

$$T(y, \mathbf{x}) = \frac{1.68}{0.837} = 2.007\,.$$

From the hint we know that a statistic value of 1.978 would lead to a p-value of $2 \cdot 0.0249 < 0.05$. Since $2.007 > 1.978$, it follows that

$$P_{H_0}(|T(Y, \mathbf{x}))| > 2.007) < P_{H_0}(|T(Y, \mathbf{x}))| > 1.978) = 2 \cdot 0.0249 < 0.05\,.$$

Hence, we can reject the null hypothesis at the significance level 0.05. The data shows enough evidence for different slops of `Bwt` for each category of `Sex`.

Problem 8 (7 credits)

Spam filters are built on principles similar to those used in logistic regression, which may be applied to estimate the probability that a message is spam (or not) on the basis of characteristics of the considered message. Here, we consider a data set whose variables are either numerical or binary indicator variables. The variable names are as follows (we avoid further details for brevity):

```
names(email)
##  [1] "spam"         "to_multiple" "from"        "cc"           "sent_email"
##  [6] "image"        "attach"      "dollar"      "winner"       "inherit"
## [11] "viagra"       "password"    "num_char"    "line_breaks"  "format"
## [16] "re_subj"      "exclaim_subj" "urgent_subj" "exclaim_mess" "number"
```

We begin by splitting the data into a training set `email_train` and a test set `email_test`. We then fit a logistic regression model to the training data, using all available predictor variables. The values of the response are `1` (for spam) and `0` (for not spam).

```
model_full <-glm(spam~., data = email_train, family = binomial)
```

a) The full model contains uninformative predictor variables, which we want to identify by running a backward selection. In one of the steps, we obtain the following values

```
Start:    AIC=1329.96
spam ~ to_multiple + from + cc + sent_email + image + attach +
  dollar + winner + inherit + viagra + password + num_char +
  line_breaks + format + re_subj + exclaim_subj + urgent_subj +
  exclaim_mess+number

               Df Deviance    AIC
<none>              1288.0 1330.0
- viagra        1   1291.9 1331.9
- num_char      1   1292.5 1332.5
- sent_email    1   1403.1 1443.1
- inherit       1   1293.4 1333.4
- from          1   1296.3 1336.3
- format        1   1296.4 1336.4
- image         1   1299.0 1339.0
- cc            1   1288.1 1328.1
- dollar        1   1302.7 1342.7
- attach        1   1304.7 1344.7
- line_breaks   1   1305.0 1345.0
- exclaim_mess  1   1309.2 1349.2
- exclaim_subj  1   1288.2 1328.2
- urgent_subj   1   1298.1 1338.1
- re_subj       1   1310.8 1350.8
- password      1   1304.9 1344.9
- winner        1   1312.2 1352.2
- number        2   1346.2 1384.2
- to_multiple   1   1375.7 1415.7
```

Which variable should be removed (if any) in this step? Justify your answer.
Remark: Be aware that the output is not given in the standard form.

In each step we remove one of the predictor variables, if this leads to a decrease in the AIC value. We pick the variable which results in the largest decline. The AIC value of the current model is 1329.96. Removing `cc` leads to an AIC value of 1328.1, which is the smallest value given in the output. Hence, `cc` will be removed in this step.

b) After some additional steps and further modifications we end up with the following model

0
1
2

```
tidy(red_model)
## # A tibble: 5 x 5
##   term         estimate std.error statistic  p.value
##   <chr>           <dbl>     <dbl>     <dbl>    <dbl>
## 1 (Intercept)  -1.13     0.0849      -13.3  1.56e-40
## 2 to_multiple1 -2.17     0.329        -6.62 3.60e-11
## 3 winneryes     2.02     0.369         5.48 4.31e- 8
## 4 line_breaks  -0.00436  0.000494     -8.83 1.05e-18
## 5 re_subj1     -3.03     0.387        -7.82 5.08e-15
```

containing the predictor variables: `to_multiple` (`0`=no, `1`=yes), `winner` (`yes`/`no` term winner contained in email), the number of `line_breaks` and `re_subj` (`1`= subject started with "Re:", `0`= subject didn't started with "Re:").
Give an **interpretation** of the estimated slopes of `line_breaks` and `re_subj` in terms of the odds ratio.

Remark: Computing the odds ratio is necessary, but not sufficient. You should describe in words the meaning of the computed value.

The odds ratio is equal to $e^{-0.00436} \approx 0.996$for each additional line break, i.e. that the odds of being a spam email is decreased by a bit more than 0.4% if the email contains an additional line break, given all else being equal.
The odds ratio is equal to $e^{-3.027} = 0.048$when the subject of the email starts with "Re:" compared to not, i.e. that the odds of being a spam email is decreased by almost 95.2% if the subject of the email starts with "Re:", given all else being equal.

c) Given the full model, we are now able to compute predictions for the test dataset.

0
1
2

```
email_test$pred_prob <-predict(model_full, newdata = email_test, type = "response")
```

To classify an email as spam or not, we need to threshold the estimated probabilities of being spam. Choosing a threshold of 0.5 leads to the following table

```
##           Truth
## Prediction FALSE TRUE
##      FALSE   706   64
##      TRUE      5   10
```

Compute sensitivity and specificity of the predictions.

The sensitivity is defined as $\frac{\#TP}{\#TP+\#FN}$ and for this prediction equal to

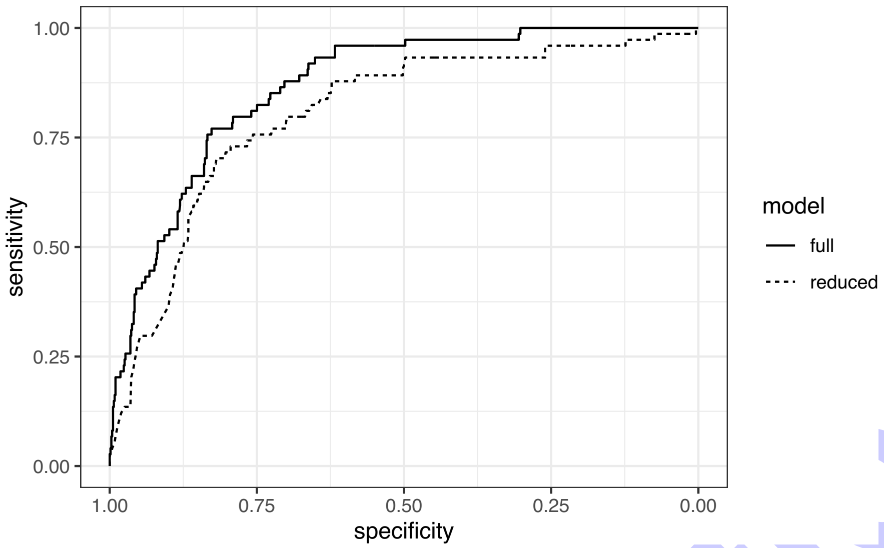
$$\frac{10}{10+64} = 0.135.$$

The specificity is defined as $\frac{\#TN}{\#TN+\#FP}$ and for this prediction equal to

$$\frac{706}{706+5} = 0.993.$$

0
1
2

d) The following graphic contains the ROC curves of the full and reduced model.



The AUC value of the full model is equal to 0.867. Decide if the reduced model has a smaller or larger AUC value. Justify your answer.

The AUC value is the area under the ROC curve. Since the ROC curve of the full model is always above the ROC curve of the reduced model, the AUC value of the reduced model will be **smaller** than 0.867.

Additional space for solutions–clearly mark the (sub)problem your answers are related to and strike out invalid solutions.

Cheat sheet

Exploring data:

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \qquad s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2 \qquad s = \sqrt{s^2} \qquad IQR = Q_3 - Q_1 \qquad Q_1 - 1.5\cdot IQR, \quad Q_3 + 1.5\cdot IQR$$

$$r_{(x,y),n} = \frac{\frac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})}{s_x \cdot s_y}$$

Probability:

$P(A^c) = 1 - P(A)$; $P(A\cap B) = P(A)\cdot P(B)$, A, B independent; $P(A\cup B) = P(A) + P(B) - P(A\cap B)$

$P(A|B) = \frac{P(A\cap B)}{P(B)}$; $E(X) = \sum_{i=1}^{k} x_i P(X = x_i)$, $E(X) = \int_{\mathbb{R}} x\cdot f(x)dx$; $Var(X) = E(X^2) - E(X)^2$;

$SD(X) = \sqrt{Var(X)}$; $Cov(X, Y) = E(X\cdot Y) - E(X)\cdot E(Y)$; $Z = \frac{X - E(X)}{\sqrt{Var(X)}}$

Geom. dist.: $P(X = x) = (1-p)^{x-1}p$, $x\in\mathbb{N}$; $E(X) = \frac{1}{p}$, $SD(X) = \sqrt{\frac{1-p}{p^2}}$

Binom. dist.: $P(X = x) = \frac{n!}{x!(n-x)!}p^x(1-p)^{n-x}$, $x\in\{0,1,2,...,n\}$; $E(X) = np$, $SD(X) = \sqrt{np(1-p)}$

Neg. Binom. dist.: $P(X = x) = \frac{(x-1)!}{(k-1)!(x-k)!}p^k(1-p)^{x-k}$, $x \geq k$

Pois. dist.: $P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}$, $x\in\mathbb{N}_0$; $E(X) = \lambda$, $SD(X) = \sqrt{\lambda}$

Inference for categorical data:

$$\hat{p} \pm z^*\cdot SE(\hat{p}) \qquad SE(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \qquad Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

$$(\hat{p}_1 - \hat{p}_2) \pm z^*\cdot SE(\hat{p}_1 - \hat{p}_2) \qquad SE(\hat{p}_1 - \hat{p}_2) = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\hat{p}_{pool}(1-\hat{p}_{pool})\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}} \qquad \hat{p}_{pool} = \frac{\hat{p}_1 n_1 + \hat{p}_2 n_2}{n_1 + n_2}$$

$\chi^2 = \sum \frac{(observed - expected)^2}{expected}$, $expected_{ij} = \frac{\text{row } i \text{ total}\cdot\text{column } j \text{ total}}{\text{table total}}$ (ind.), $df = k-1$ or $(r-1)\cdot(c-1)$

Inference for numerical data:

$$\bar{x} \pm t^*_{df}\cdot SE(\bar{x}) \qquad T_{df}(\mathbf{x}) = \frac{\bar{x} - \mu_0}{SE(\bar{x})} \qquad SE(\bar{x}) = \frac{s(x)}{\sqrt{n}} \qquad df = n-1$$

$$(\bar{x}-\bar{y}) \pm t^*_{df}\cdot SE(\bar{x}-\bar{y}) \qquad T_{df}(\mathbf{x},\mathbf{y}) = \frac{(\bar{x}-\bar{y}) - 0}{SE(\bar{x}-\bar{y})} \qquad SE(\bar{x}-\bar{y}) = \sqrt{\frac{s(x)^2}{n_1} + \frac{s(y)^2}{n_2}} \qquad df = \min(n_1-1, n_2-1)$$

two-samples: $n \geq \frac{2\cdot\sigma_0^2}{\delta_0^2}(z_{1-\alpha^*} + z_{1-\beta})^2$; one-sample: $n \geq \frac{\sigma_0^2}{\delta_0^2}(z_{1-\alpha*} + z_{1-\beta})^2$ with $\alpha^* \in \{\alpha, \alpha/2\}$

ANOVA and linear regression:

$F = \frac{MSG}{MSE}$ $df_1 = k-1$, $df_2 = n-k$, for k groups; $SSG(\mathbf{y}) = \sum_{i=1}^{k} n_i(\bar{y}_i - \bar{y})^2$, $SST(\mathbf{y}) = \sum_{j=1}^{n}(y_j - \bar{y})^2$,

$SSE(\mathbf{y}) = SST(\mathbf{y}) - SSG(\mathbf{y})$; $p_{adj,i} = K\cdot p_i$

$$\hat{y} = b_0 + b_1 x \qquad b_1 = r_{(x,y),n}\cdot\frac{s_y}{s_x} \qquad \bar{y} = b_0 + b_1\bar{x} \qquad \text{residual} = y - \hat{y} \qquad \hat{\sigma} = \sqrt{\frac{1}{n-2}\sum(y_i - \hat{y}_i)^2}$$

$$T_{df}(\mathbf{y},\mathbf{x}) = \frac{b_1(\mathbf{y},\mathbf{x}) - \beta_{1,0}}{SE_{b_1}(\mathbf{y},\mathbf{x})} \qquad b_1(\mathbf{y},\mathbf{x}) \pm t^*_{df}\cdot SE_{b_1}(\mathbf{y},\mathbf{x}) \qquad df = n-2; \qquad R^2 = \frac{\sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

$$R^2_{adj} = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2/(n-k-1)}{\sum_{i=1}^{n}(y_i - \bar{y})^2/(n-1)}; \qquad AIC = \text{constant} + n\cdot\ln\left(\sum_{i=1}^{n} e_i^2\right) + 2k$$

Logistic regression:

$$\text{logit}(p_i) = \log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \beta_1 x_{1,i} + \cdots + \beta_k x_{k,i} \qquad OR_j = \frac{\text{odds when } x_{i,j} = x+1}{\text{odds when } x_{i,j} = x} = e^{\beta_j}$$

sensitivity = P(Test + | Condition +), specificity = P(Test − | Condition −)

Additional space for solutions–clearly mark the (sub)problem your answers are related to and strike out invalid solutions.